

Zhimeng Zhu

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EDUCATION

Huazhong University of Science and Technology (HUST)

09/2022-06/2026

- Bachelor of Engineering in Automation
- Scholarship: Social Service Scholarship (Top 10%, 09/2024)

GPA: 3.18/5.0

PUBLICATIONS

[1] **Zhu, Zhimeng.** (2025). Enhancing Brain Tumor Detection: A Comparative Study of CNN Architectures Using MRI Data. *ITM Web of Conferences*. 70. 03014. DOI:[10.1051/itmconf/20257003014](https://doi.org/10.1051/itmconf/20257003014).

RESEARCH EXPERIENCES

Lab: National Key Laboratory of Multispectral Information Intelligent Processing Technology

Advisor: Assoc. Prof. Tian Tian, School of Artificial Intelligence and Automation, HUST

Multispectral Object Detection and Embedded Deployment

09/2024-Present

- Designed a deep learning-based multispectral object detection model using TensorFlow and PyTorch to enhance accuracy and robustness under complex lighting and environmental conditions
- Built a high-quality multispectral dataset through data collection, cleaning, and precise annotation for model training and validation
- Optimized model architecture and training strategies using data augmentation, residual connections, and batch normalization to mitigate overfitting and vanishing gradient problems
- Deployed the optimized model on an NVIDIA Jetson Xavier NX embedded platform, applying model quantization and pruning to achieve real-time inference and efficient on-device performance

First Prize, 20th "Chuanshibao Cup" Transportation Science and Technology Competition (University-Level)

Advisor: Dr. Zehao Jiang, School of Civil and Hydraulic Engineering, HUST

Intelligent Q&A System for Transportation Engineering Standards Based on DeepSeek

03/2024-05/2025

- Developed a domain-specific large language model for transportation engineering standards to address inefficiencies caused by redundant, inconsistent, and incomplete standard systems
- Processed national and industry standard documents using a legal document processing pipeline for batch JSON structuring, data cleaning, and normalization
- Built the base model on DeepSeek-R1 (32B parameters), applying domain fine-tuning by constructing a learning database and inputting lexicon, improving terminology recognition accuracy to $F1 = 0.89$
- Designed an output control mechanism to ensure concise and professional answers, automatically citing standard references and section numbers to provide low-cost, high-precision information services
- Constructed a standards knowledge graph enabling conflict detection and redundancy identification, supporting intelligent Q&A and dynamic optimization of the standards system

Lab: Key Laboratory of Image Processing and Intelligent Control

Project Leader, Advisor: Assoc. Prof. Lintao Ye, School of Artificial Intelligence and Automation, HUST

Federated Learning for Personalized Financial Services

04/2024-04/2025

- Applied federated learning to personalized financial services to enhance recommendation accuracy and client risk assessment while ensuring data privacy
- Reproduced core algorithms (FedAvg and FedProx) and built a distributed training framework in PyTorch using the Federated-Learning-PyTorch dataset to simulate decentralized financial environments
- Addressed domain-specific Non-IID challenges such as cross-bank customer data, regional consumer behavior, and temporal transaction patterns through algorithmic tuning
- Integrated the FedProx regularization term to mitigate client-side heterogeneity, improving model stability and generalization in imbalanced real-world financial data scenarios

Online Research Seminar: Machine Learning and Data Science - Development of Applications

Team Leader, Advisor: Prof. Mark Vogelsberger, Department of Physics, Massachusetts Institute of Technology

Enhancing Brain Tumor Detection: A Comparative Study of CNN Architectures Using MRI Data

07/2024-08/2024

- Proposed an enhanced CNN model, VGG19-BMT (VGG19-Brain's MRI for Tumor), by adjusting convolutional layers and feature extraction modules for improved brain tumor detection
- Preprocessed the open-access Kaggle Brain MRI dataset with normalization and data augmentation (rotation, brightness shift, scaling) to enhance model generalization
- Applied ReduceLROnPlateau for dynamic learning rate adjustment and incorporated Dropout and a custom classifier head to prevent overfitting on limited medical data
- Trained VGG19-BMT with cross-entropy loss and compared it with VGG16, VGG19, ResNet18, and EfficientNet-B2 under consistent settings

- Visualized core performance metrics (accuracy, precision, recall, F1-score) to assess convergence behavior and model reliability
- Confirmed that VGG19-BMT outperformed all baselines, achieving 99.51% accuracy and 0.9498 F1-score, with potential applicability to other medical imaging tasks like lung nodule or breast cancer screening

Image Generation with Variational Autoencoders: Model Optimization and Improvement Exploration

06/2024-08/2024

- Optimized the encoder, decoder, and custom loss function (reconstruction loss + KL divergence) of the standard VAE (Variational Autoencoder) model, and developed three variants (BetaVAE, Conditional VAE, InfoVAE) to improve the quality-diversity trade-off in image generation
- Preprocessed the CelebA dataset (over 200,000 face images) via data cleaning, normalization, and augmentation (rotation, flipping, and cropping) to enhance model robustness and generalizability
- Implemented all models in PyTorch and conducted GPU-based training, tuning learning rates, batch sizes, and optimizer settings to ensure stable convergence and high-quality outputs
- Evaluated models' performance using SSIM, and conducted user studies to incorporate human-perceived image quality in final assessments
- Identified InfoVAE as the best-performing variant, achieving the highest SSIM (0.338), with strong retention of semantic details and structure, offering insights into VAE-based generative architecture design

SELECTED PROJECTS

Cross-Domain Few-Shot Object Recognition

04/2025-06/2025

- Implemented Prototypical Network (baseline), Improved ProtoNet (feature normalization and attention mechanism), and Matching Network (contextual embedding and support set weighting) on the Office-Home dataset, exploring transferability and generalization in cross-domain few-shot image recognition
- Developed models in PyTorch and applied target-domain fine-tuning, significantly improving accuracy in cross-domain few-shot and 1-shot classification tasks
- Designed multi-source domain adaptation experiments based on transfer learning and few-shot learning principles to evaluate model performance under different domain shifts
- Compared model robustness across domain combinations, demonstrating that Improved ProtoNet achieved the best performance in most scenarios, with particularly strong results in 1-shot tasks

Autonomous Water Quality Monitoring System Based on Unmanned Surface Vehicle

01/2023-04/2023

- Designed an intelligent monitoring system integrating Unmanned Surface Vehicle (USV) autonomous navigation with real-time data analysis to support ecological protection and sustainable lake management
- Optimized USV path planning using A* and Dijkstra algorithms, adapting to complex lake topography and variable hydrodynamic conditions
- Implemented statistical analysis and machine learning techniques to preprocess water quality data and develop predictive models for key environmental indicators
- Applied environmental science principles and fuzzy mathematics to improve model accuracy, system robustness, and monitoring efficiency

ADDITIONAL INFORMATION

Computer Skills: Python, C, C++, MATLAB, Java, R

Interests: Piano (13 years; multiple awards, including a provincial-level prize), Basketball